

Problem 5

Consider a simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (i = 1, \dots, n)$$

where $x_1, \dots, x_n$ are constants, α, β are parameters, $\varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$ ($i = 1, \dots, n$)

Suppose $n = 3$ and three data points: $(-1, y_1), (0, y_2), (1, y_3)$

- (a) Find the design matrix X
- (b) Express the least squares estimators of β_0, β_1 in terms of y_1, y_2, y_3
- (c) Find out the expression of the regression sum of squares in terms of y_1, y_2, y_3
- (d) Suppose $y_3 = y_1 + 2$ and the total sum of squares
 $SST := S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = 2.5$
What is the value of the coefficient of determination R^2 ?
Does the regression line fit the three data points well?
- (e) Under the condition of (d), find the 95% confidence intervals for β_0, β_1 respectively.

原题截图:

5. Consider a simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{with } \varepsilon_i \sim N(0, \sigma^2) \quad i = 1, 2, \dots, n$$

and $\varepsilon_1, \dots, \varepsilon_n$ are independent. We have three data points: $(-1, y_1), (0, y_2), (1, y_3)$.

- (a) Find the design matrix X .
- (b) Express the least squares estimates of β_0 and β_1 in terms of y_1, y_2 and y_3 .
- (c) Find out the expression of the regression sum of squares in terms of y_1, y_2 and y_3 .
Suppose now we know that $y_3 = y_1 + 2$ and the total corrected sum of squares
 $S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = 2.5$.
- (d) What is the value of the coefficient of determination r^2 ? Does the regression line fit the three data points well?
- (e) Find the 95% confidence intervals for β_0 and β_1 respectively.

Part (a)

Find the design matrix X

Solution:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

于是 $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ ($i = 1, 2, 3$) 可以写成紧凑形式 $y = X\beta + \varepsilon$ (其中 $\beta = [\beta_0, \beta_1]^T$)

Part (b)

Express the least squares estimators of β_0, β_1 in terms of y_1, y_2, y_3

Solution:

$$\begin{aligned}
\hat{\beta} &:= \arg \min_{\beta \in \mathbb{R}^2} \|y - X\beta\|_2^2 \\
&= \text{solution of } \{-2X^T(y - X\beta) = 0_2\} \\
&= (X^T X)^{-1} X^T y \\
&= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{3}(y_1 + y_2 + y_3) \\ \frac{1}{2}(y_3 - y_1) \end{bmatrix}
\end{aligned}$$

因此我们有:

$$\begin{cases} \hat{\beta}_0 = \frac{1}{3}(y_1 + y_2 + y_3) = \bar{y} \\ \hat{\beta}_1 = \frac{1}{2}(y_3 - y_1) \end{cases}$$

Part (c)

Find out the expression of the regression sum of squares in terms of y_1, y_2, y_3

Solution:

$$\begin{aligned}
\text{SSR} &= \|\hat{y} - \bar{y}1_3\|_2^2 \\
&= \|X\hat{\beta} - \frac{1}{3}1_3^T y 1_3\| \quad (\text{note that } \hat{\beta} = (X^T X)^{-1} X^T y) \\
&= \| [X(X^T X)^{-1} X^T - \frac{1}{3}1_3 1_3^T] y \|_2^2 \\
&= y^T [X(X^T X)^{-1} X^T - \frac{1}{3}1_3 1_3^T] y \\
&= y^T \left(\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}^T - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \left(\begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \\ & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \left(\begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) y \\
&= y^T \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix} y \\
&= \frac{1}{2}(y_1 - y_3)^2
\end{aligned}$$

Part (d)

Suppose $y_3 = y_1 + 2$ and the total sum of squares $\text{SST} := S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = 2.5$

What is the value of the coefficient of determination R^2 ?

Does the regression line fit the three data points well?

Solution:

由于 $y_3 = y_1 + 2$, 故 $SSR = \frac{1}{2}(y_1 - y_3)^2 = \frac{1}{2} \cdot 2^2 = 2$

因此 $R^2 = \frac{SSR}{SST} = \frac{2}{2.5} = 0.8$

这个数值很接近于 1, 故模型较好地拟合了 3 个数据点.

Part (e)

Under the condition of (d), find the 95% confidence intervals for β_0, β_1 respectively.

Solution:

根据 $\begin{cases} y_3 = y_1 + 2 \\ S_{yy} = \sum_{i=1}^3 (y_i - \bar{y})^2 = y_1^2 + y_2^2 + y_3^2 - 3\bar{y}^2 = 2.5 \text{ 我们可以得到:} \\ \bar{y} = \frac{1}{3}(y_1 + y_2 + y_3) \end{cases}$

$$\begin{aligned} 2.5 &= y_1^2 + y_2^2 + (y_1 + 2)^2 - 3 \cdot \frac{1}{9}(2y_1 + y_2 + 2)^2 \\ &= 2y_1^2 + y_2^2 + 4y_1 + 4 - \frac{1}{3}(4y_1^2 + y_2^2 + 4 + 4y_1y_2 + 8y_1 + 4y_2) \\ &= \frac{2}{3}y_1^2 + \frac{2}{3}y_2^2 + \frac{4}{3}y_1 - \frac{4}{3}y_2 - \frac{4}{3}y_1y_2 + \frac{8}{3} \\ &= \frac{2}{3}(y_1^2 + y_2^2 + 1 + 2y_1 - 2y_2 - 2y_1y_2) + 2 \\ &= \frac{2}{3}(y_1 - y_2 + 1)^2 + 2 \end{aligned}$$

$$(y_1 - y_2 + 1)^2 = \frac{3}{4} \Rightarrow (y_1 - y_2) = -1 \pm \frac{\sqrt{3}}{2}$$

但遗憾的是, 根据已有数据我们仍然无法计算出 $\hat{\beta}_0 = \bar{y}$

根据 Part (b) 我们有:

$$\begin{aligned} \hat{\beta} &:= \arg \min_{\beta \in \mathbb{R}^2} \|y - X\beta\|_2^2 \\ &= \text{solution of } \{-2X^T(y - X\beta) = 0_2\} \\ &= (X^T X)^{-1} X^T y \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3}(y_1 + y_2 + y_3) \\ \frac{1}{2}(y_3 - y_1) \end{bmatrix} \quad (\text{note that } y_3 = y_1 + 2) \\ &= \begin{bmatrix} \bar{y} \\ 1 \end{bmatrix} \end{aligned}$$

因此我们有:

$$\begin{aligned}
\hat{\beta} &= (X^T X)^{-1} X^T y \\
&= (X^T X)^{-1} X^T (X\beta + \varepsilon) \\
&= \beta + (X^T X)^{-1} X^T \varepsilon \\
\hline
\hat{\beta} - \beta &= (X^T X)^{-1} X^T \varepsilon \\
&\sim N((X^T X)^{-1} X^T 0_n, (X^T X)^{-1} X^T \cdot \sigma^2 I_n \cdot [(X^T X)^{-1} X^T]^T) \\
&= N(0_2, \sigma^2 (X^T X)^{-1}) \\
&= N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \sigma^2 \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1}\right) \\
&= N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{3}\sigma^2 & 0 \\ 0 & \frac{1}{2}\sigma^2 \end{bmatrix}\right)
\end{aligned}$$

而 σ^2 的无偏估计量 $s^2 = \frac{1}{n-2} \text{SSE} = \frac{1}{n-2} (\text{SST} - \text{SSR}) = \frac{1}{3-2} (2.5 - 2) = 0.5 \sim \sigma^2 \chi^2_{(1)}$

因此用于对 β_0, β_1 进行区间估计的枢轴量为:

$$\begin{aligned}
\frac{\hat{\beta}_0 - \beta_0}{\sqrt{\frac{1}{3}s^2}} &= \sqrt{6}(\hat{\beta}_0 - \beta_0) \sim t_1 \\
\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\frac{1}{2}s^2}} &= 2(\hat{\beta}_1 - \beta_1) \sim t_1
\end{aligned}$$

因此 β_1 的 95% 置信区间为 $[1 - \frac{1}{2}t_{1,0.975}, 1 + \frac{1}{2}t_{1,0.975}]$

其中 $t_{1,0.975} \approx 12.71$ 为自由度为 1 的 t 分布的 $1 - \frac{1}{2}(1 - 0.95) = 0.975$ 分位数.

但是由于 $\hat{\beta}_0 = \bar{y}$ 计算不出来, 因此无法给出 β_0 的区间估计.

如果真要做的话, 似乎只能对 $(y_1 - y_2) = -1 \pm \frac{\sqrt{3}}{2}$ 分类讨论,

得到 $\bar{y}$ 关于 y_1 的两种表达式, 分别给出置信区间.

我在想, 老师您是不是遗漏了某个条件呢, 例如 " $\bar{y} = 0.5$ " 之类的?